

Set 14: Survey Results Analysis

DSA – Data Handling and Visualization | Lab Experiment

R.No	Reg. No	Name
14	192524098	Divyasree Muthukumar

Aim

To analyse survey responses using a stacked bar chart for Question 1, a radar chart showing the overall answer pattern, and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
survey <- data.frame(
  Respondent = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
write.csv(survey, "Survey_Results.csv", row.names = FALSE) # for Tableau
pal <- c(A = "#0072B2", B = "#E69F00", C = "#009E73", D = "#CC79A7")
```

Q1. Stacked Bar Chart – Distribution of Answers for Question 1

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

R Program:

```
# Q1. Stacked bar chart - distribution of answers for Question 1
q1 <- as.data.frame(table(Answer = factor(survey$Question_1, levels = c("A", "B", "C", "D"))))
names(q1)[2] <- "Responses"
q1$Percent <- round(100 * q1$Responses / sum(q1$Responses), 1)
print(q1, row.names = FALSE)

p_stk <- ggplot(q1[q1$Responses > 0, ], aes(x = "Question 1", y = Responses, fill = Answer)) +
  geom_col(width = 0.45, colour = "white") +
  geom_text(aes(label = paste0(Answer, ": ", Responses, " (", Percent, "%)"),
    position = position_stack(vjust = 0.5), colour = "white",
    fontface = "bold", size = 4) +
  scale_fill_manual(values = pal) +
  scale_y_continuous(breaks = 0:3) +
  labs(title = "Distribution of Answers for Question 1",
    x = NULL, y = "Number of Responses", fill = "Answer") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_stk)
ggsave("s14_q1_stacked.png", p_stk, width = 6, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Answer Responses Percent
A           1      33.3
```

B	1	33.3
C	1	33.3
D	0	0.0

Output:

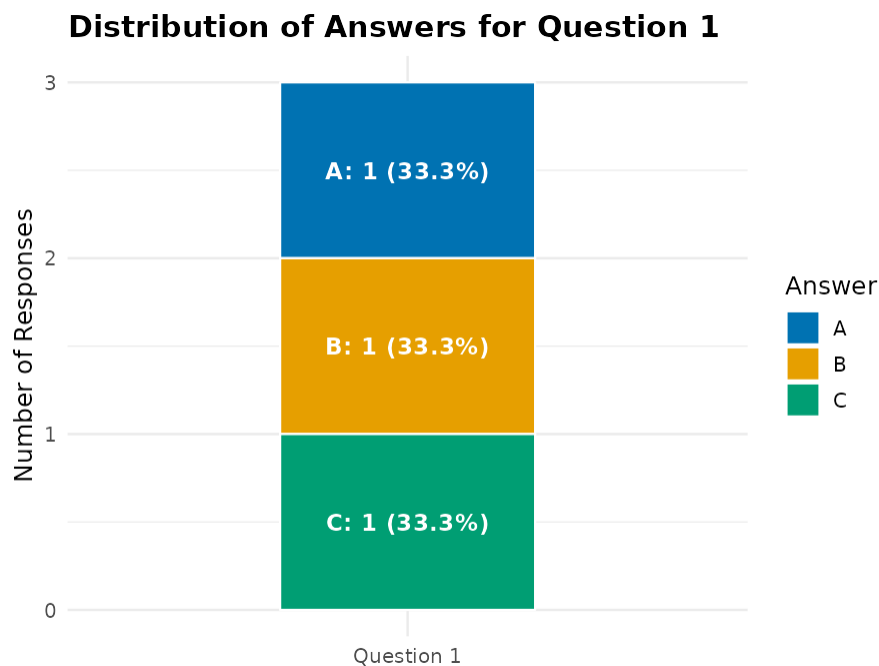


Figure 1: Stacked bar of Question 1 answers

Inference:

Question 1 answers are split evenly: A, B and C each got 1 response (33.3%), and D got none. There is no consensus on Question 1.

Q2. Radar Chart – Overall Pattern of Responses

Generate a radar chart to visualize the overall pattern of responses across all three questions.

R Program:

```
# Q2. Radar chart - overall pattern of responses across all three questions
long <- data.frame(Question = rep(c("Question 1", "Question 2", "Question 3"), each = 3),
  Answer = c(survey$Question_1, survey$Question_2, survey$Question_3))
long$Answer <- factor(long$Answer, levels = c("A", "B", "C", "D"))
cnt <- as.data.frame(table(Question = long$Question, Answer = long$Answer))
names(cnt)[3] <- "Responses"
print(xtabs(Responses ~ Question + Answer, cnt))
cat("\nOverall totals:\n"); print(table(long$Answer))

# Build a straight-edged radar: one spoke per answer option (A-D), radius = count
opts <- levels(long$Answer); k <- length(opts)
ang <- pi / 2 - 2 * pi * (seq_len(k) - 1) / k # A at top, clockwise
cnt$angle <- ang[match(cnt$Answer, opts)]
cnt$x <- cnt$Responses * cos(cnt$angle); cnt$y <- cnt$Responses * sin(cnt$angle)
grid_df <- do.call(rbind, lapply(1:3, function(r)
  data.frame(r = r, x = r * cos(c(ang, ang[1])), y = r * sin(c(ang, ang[1])))))
spokes <- data.frame(x = 0, y = 0, xend = 3 * cos(ang), yend = 3 * sin(ang))
lbl <- data.frame(Answer = opts, x = 3.45 * cos(ang), y = 3.45 * sin(ang))

p_radar <- ggplot() +
  geom_path(data = grid_df, aes(x, y, group = r), colour = "grey80") +
  geom_segment(data = spokes, aes(x, y, xend = xend, yend = yend), colour = "grey80") +
  annotate("text", x = 0.12, y = 1:3 + 0.12, label = 1:3, size = 3, colour = "grey45") +
  geom_polygon(data = cnt, aes(x, y, group = Question, fill = Question, colour = Question),
    alpha = 0.2, linewidth = 1) +
  geom_point(data = cnt, aes(x, y, colour = Question), size = 2.5) +
  geom_text(data = lbl, aes(x, y, label = paste("Option", Answer)), fontface = "bold") +
  scale_fill_manual(values = c("#0072B2", "#E69F00", "#009E73")) +
  scale_colour_manual(values = c("#0072B2", "#E69F00", "#009E73")) +
  coord_equal(xlim = c(-3.9, 3.9), ylim = c(-3.7, 3.7)) +
  labs(title = "Radar Chart: Response Pattern across Questions",
    subtitle = "Spokes = answer options; distance from centre = number of responses",
```

```

    fill = "Question", colour = "Question") +
  theme_void(base_size = 12) +
  theme(plot.title = element_text(face = "bold"), legend.position = "right")
print(p_radar)
ggsave("s14_q2_radar.png", p_radar, width = 7, height = 5.8, dpi = 150, bg = "white")

```

Console Output:

```

      Answer
Question  A B C D
Question 1 1 1 1 0
Question 2 2 1 0 0
Question 3 0 1 1 1

Overall totals:

A B C D
3 3 2 1

```

Output:

Radar Chart: Response Pattern across Questions

Spokes = answer options; distance from centre = number of responses

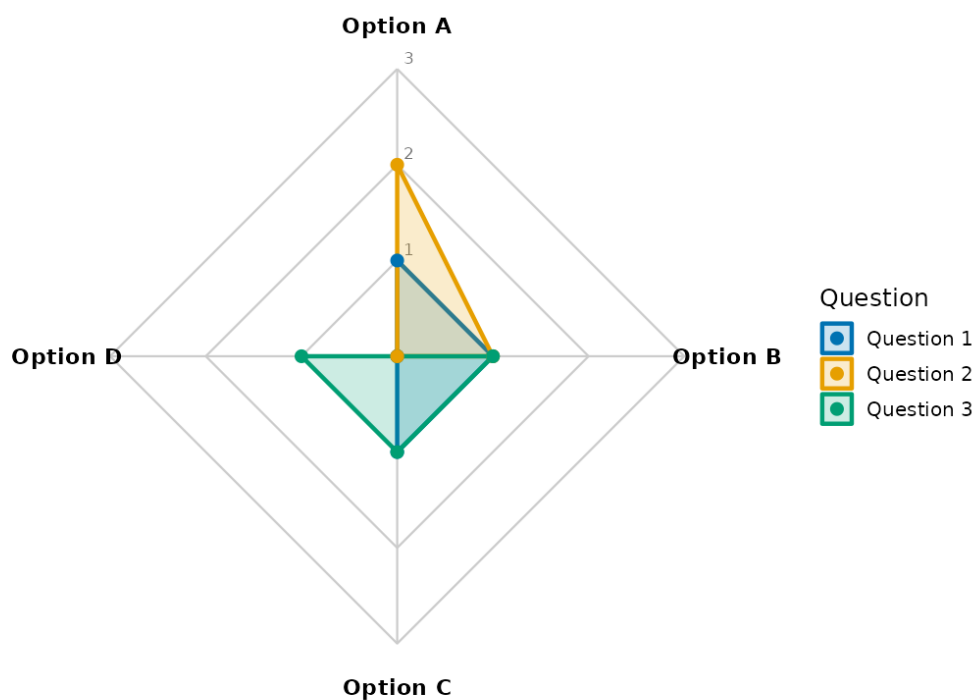


Figure 2: Radar chart – one polygon per question across answer options A–D

Inference:

Each spoke is an answer option and each coloured polygon is a question.

Question 2 stretches furthest toward **Option A** (2 responses), so it has the clearest preference. Question 1 forms an even triangle over A, B and C. Question 3 leans toward B, C and D and is the only question where D was chosen.

Overall, A and B are the most frequent answers (3 each), followed by C (2) and D (1).

Q3. Table – Survey Response Data

Build a table to show the survey response data for each respondent. Label the table elements.

R Program:

```
# Q3. Labelled table of survey responses
tbl <- survey
names(tbl) <- c("Respondent", "Question 1", "Question 2", "Question 3")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Survey Response Data by Respondent",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s14_q3_table.png", width = 900, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

```
Respondent Question 1 Question 2 Question 3
1           A         B         C
2           B         A         D
3           C         A         B
```

Output:

Survey Response Data by Respondent

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 3: Labelled survey response table

Inference:

The table shows each respondent's answers to the three questions. All three respondents gave different answer patterns.

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

R Program:

```
# Q4. Dashboard layout preview (Tableau dashboard built from Survey_Results.csv)
library(patchwork)
dash <- (p_stk | p_radar) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.5, 0.5), widths = c(0.8, 1.2)) +
  plot_annotation(title = "Survey Results Dashboard",
    theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s14_dashboard.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

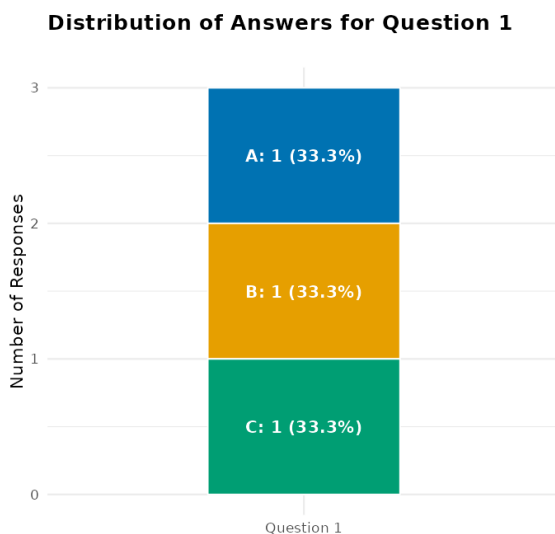
Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Survey_Results.csv**. In the Data Source tab, select Question_1, Question_2 and Question_3 → right-click → **Pivot**. Rename the pivot fields to **Question** and **Answer**.
- 2 **Sheet 1 – "Q1 Stacked Bar"**: filter Question = Question_1. Drag Question to Columns and **CNT(Respondent)** to Rows, then drag Answer to **Color** and Label. Add a Percent of Total quick table calculation to the label.

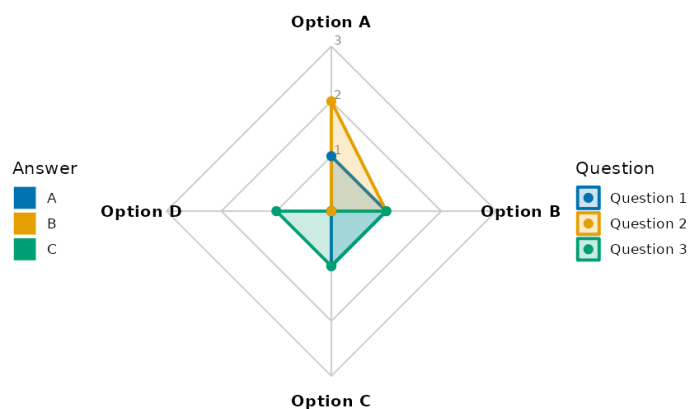
- Sheet 2 – "Radar":** Tableau has no built-in radar chart, so create these calculated fields:
 $Angle = \text{CASE } [Answer] \text{ WHEN "A" THEN 0 WHEN "B" THEN 1 WHEN "C" THEN 2 ELSE 3 END } * 2 * \pi / 4$
 $X = \text{COUNT}([Respondent]) * \sin(\text{MIN}([Angle]))$ and $Y = \text{COUNT}([Respondent]) * \cos(\text{MIN}([Angle]))$
Put X on Columns and Y on Rows, set Marks to **Polygon** (or Line), drag Question to Color, and put Answer on Path (sorted A→D) and Label.
- Sheet 3 – "Response Table":** drag Respondent to Rows, Question to Columns and Answer to Text, which gives a crosstab matching the original data.
- From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- Drag the sheets onto the canvas: Q1 Stacked Bar (top-left), Radar (top-right), Response Table (bottom). Add a **Text** object at the top for the dashboard title.
- On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Survey_Results) so one filter controls every view.
- Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.

Dashboard Layout:

Survey Results Dashboard



Radar Chart: Response Pattern across Questions
Spokes = answer options; distance from centre = number of responses



Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Clicking an answer segment in the stacked bar highlights the matching respondent in the table and that option on the radar, which links individual responses to the overall pattern.

Result

The survey results were visualised with a stacked bar chart, a radar chart and a labelled table in R, and combined into an interactive Tableau dashboard. Question 1 is evenly split across A, B and C; Question 2 favours A; and Options A and B are the most common answers overall.